

DUTCH HARBOR, AK, USA

WMO: 704890

Lat: 53.895N

Lon: 166.543W

Elev: 3

StdP: 101.29

Time Zone: -9.00 (NAK)

Period: 94-19

WBAN: 25616

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-7.1	-5.6	-12.2	1.3	-5.8	-10.9	1.5	-4.7	17.1	2.9	15.4	1.3	6.8	330	0.796

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	4.2	17.5	14.3	15.9	13.2	14.1	11.6	14.7	17.1	13.3	15.6	12.1	13.9	4.9	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
13.0	9.3	16.5	12.0	8.8	15.2	10.9	8.1	13.8	40.8	17.1	37.3	15.6	34.2	14.1	20.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	15.1	12.9	11.5	DB	-8.2	21.5	3.0	2.1	-10.3	23.0	-12.0	24.2	-13.7	25.4	-15.9	27.0
(5)				WB	-9.2	17.6	2.6	1.6	-11.0	18.8	-12.6	19.8	-14.0	20.7	-15.9	21.9

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	5.4	1.0	1.4	1.3	2.8	5.2	8.2	10.8	11.9	9.7	6.4	3.7	1.9
	DBStd	4.46	2.94	2.87	2.85	2.26	1.90	1.56	1.72	2.08	1.98	2.25	2.35	2.76
	HDD10.0	1807	280	241	269	217	148	58	10	4	27	113	188	252
	HDD18.3	4730	538	474	528	467	406	304	235	201	258	370	438	511
	CDD10.0	120	0	0	0	0	0	4	33	62	19	2	0	0
	CDD18.3	1	0	0	0	0	0	0	0	1	0	0	0	0
	CDH23.3	1	0	0	0	0	0	0	0	1	0	0	0	0
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	5.0	5.3	5.5	5.3	5.4	4.6	3.9	3.6	3.8	5.0	5.6	5.9	5.7
Precipitation	PrecAvg	1761	208	172	147	95	112	66	66	76	149	193	185	225
	PrecMax	2194	429	368	374	171	259	103	185	157	253	458	497	481
	PrecMin	1260	57	30	39	37	29	18	5	12	28	110	70	48
	PrecStd	271	90	79	93	39	53	27	40	37	69	79	109	103
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.4	8.2	9.0	9.0	11.4	15.3	19.1	21.1	17.2	13.0	11.1	9.1
		MCWB	4.9	5.5	6.7	6.8	8.2	11.8	15.8	17.3	14.5	10.5	8.3	6.7
	2%	DB	6.2	6.8	6.9	7.5	9.3	12.6	16.7	18.0	15.0	11.5	9.1	7.7
		MCWB	4.2	4.5	4.5	4.9	6.5	9.8	13.6	15.0	12.6	9.0	7.0	5.7
	5%	DB	5.2	5.8	5.8	6.6	8.5	11.3	14.5	16.5	13.6	10.5	8.0	6.5
		MCWB	3.5	3.8	3.6	4.1	6.1	8.8	12.0	13.7	11.2	8.1	5.9	4.4
	10%	DB	4.3	4.7	4.7	6.0	7.7	10.4	13.1	14.9	12.6	9.4	7.2	5.4
		MCWB	2.7	3.0	2.8	3.7	5.5	8.2	10.9	12.5	10.1	7.2	5.1	3.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.4	5.8	6.9	6.9	8.4	12.4	16.2	17.4	14.8	10.9	8.5	7.3
		MCDB	7.0	7.7	8.8	8.7	10.8	14.9	19.0	20.4	16.9	12.8	10.4	8.9
	2%	WB	4.6	4.9	4.8	5.3	7.3	10.2	13.8	15.3	12.8	9.3	7.6	6.0
		MCDB	6.2	6.5	6.6	7.2	8.7	12.2	16.3	17.8	14.8	11.2	9.2	7.5
	5%	WB	3.9	4.0	3.9	4.6	6.5	9.1	12.3	14.0	11.4	8.3	6.4	4.8
		MCDB	5.4	5.5	5.5	6.4	8.1	10.9	14.2	16.4	13.3	10.0	7.9	6.5
	10%	WB	3.1	3.2	3.0	3.9	5.7	8.3	11.2	12.8	10.4	7.6	5.3	3.9
		MCDB	4.3	4.5	4.6	5.6	7.5	9.9	12.9	14.8	12.3	9.2	7.1	5.4
Mean Daily Temperature Range	5% DB	MDBR	3.2	3.5	3.4	3.3	3.4	3.5	4.1	4.2	3.5	3.2	3.4	3.3
		MCDBR	3.9	4.9	4.5	4.3	4.7	5.3	6.8	6.8	5.1	4.1	4.6	4.4
		MCWBR	3.4	3.9	3.8	3.4	3.3	3.5	4.8	4.9	4.0	3.5	4.2	3.8
	5% WB	MCDBR	3.9	4.1	4.3	4.0	4.1	4.8	6.7	6.6	4.9	3.8	4.4	4.4
		MCWBR	3.5	3.7	3.9	3.4	3.1	3.4	5.0	5.0	4.0	3.5	4.3	3.8
Clear-Sky Solar Irradiance	taub	0.310	0.346	0.354	0.402	0.400	0.393	0.392	0.393	0.375	0.353	0.334	0.305	
	taud	2.244	2.215	2.256	2.148	2.238	2.346	2.407	2.399	2.360	2.294	2.185	2.202	
	Ebn at Noon	633	727	818	820	846	857	850	827	791	712	579	549	
	Edh at Noon	62	88	105	133	129	117	108	103	95	81	65	53	
All-Sky Solar Radiation	RadAvg	0.55	1.16	2.33	3.48	4.20	4.41	3.85	3.33	2.56	1.59	0.72	0.41	
	RadStd	0.06	0.10	0.23	0.27	0.36	0.33	0.42	0.25	0.18	0.11	0.07	0.04	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	+1.60	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (1 neighbor)	N/A	N/A	+1.60	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page